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Research article

Life cycle assessment and water footprint evaluation of crude steel production: A case study in China

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ABSTRACT

China, as the world's largest crude steel producer, is suffering from water scarcity and pollution. However, only a few systematic analyses on the environmental burdens and improvements of China's crude steel production have been conducted. Therefore, it is important for research to be done how China's steel industry can be improved in environment management. To help decision-makers understand this, a life cycle water footprint analysis including gray and blue water was performed based on the methodology prescribed in the ISO 14046 standard. A life cycle assessment was also conducted to improve the environmental performance of the steel industry. Results of these assessments revealed that gray water footprint, which is mainly derived from aquatic eutrophication, carcinogens, and non-carcinogens, is higher than blue water footprint. Optimizing indirect processes, including iron ore mining, magnesium oxide production, transportation, and electricity generation, played dominant roles in the reduction of gray water footprint. Furthermore, COD, Cr (VI), phosphate, BOD_5, Hg, As, nitrogen oxides, particulates, and sulfur dioxide were the key substances for environmental improvements. The underestimation of direct water footprint showed the importance and urgency of implementing scientific and adequate monitoring indicators. Meanwhile, the environmental burden can be reduced by adopting a reasonable location of the steel industry on the basis of regional water resources and actual transportation status, improving the efficiency of raw material consumption, and optimizing the power structure.

1. Introduction

Water resource shortage is a common global problem, and the world is predicted to face about 40% global water deficit in 2030 (UNESCO, 2015). The China is also identified with water scarcity (UNESCO, 2015), and approximately 32% of the water in China faces various pollution problems (NBSC, 2016). China's iron and steel industry (ISI), which has been the world's largest crude steel production industry since 1996 (WSA, 2016), suffers from serious water resources shortage and environmental pollution problems (Gu et al., 2015; MPI, 2016). For example, 14.5% of the total wastewater and gas is produced from ISI in China (Wu et al., 2017). Moreover, the freshwater consumed by China's ISI accounts for approximately 14% of the total industrial water consumption of China (Huang et al., 2011). The wastewater discharged from ISI contains numerous toxic pollutants (e.g., heavy metals, waste oil, and cyanide) that lead to significant environmental degeneration (Gu et al., 2015). Meanwhile, ISI is highly energy intensive which caused inseparable association with water resource shortage and various pollutants emissions during its entire life cycle (Olmez et al., 2016). Those environmental impacts can be passed through the supply chain to downstream products and significantly influence the global environment (Berger et al., 2012; Wang et al., 2014). To deal with water resource problems caused by ISI, the Chinese government has enacted increasingly stringent standards to control wastewater from ISI and encourage enterprises to improve water consumption and recycling efficiency (MPI, 2016). A systematic assessment of water discharge and consumption is therefore necessary.

Water footprint (WF) is a comprehensive indicator for assessing water resource consumption and pollution caused by anthropogenic activities in geographical and temporal dimensions (Hoekstra et al., 2011). This indicator considers direct and indirect processes. To the best of our knowledge, only few analytical studies on WF have examined crude steel production. One of these limited studies is that of Gu et al. (2015), who performed a case study on ISI in Eastern China via chain summation and stepwise accumulative approach. However, this study hardly identified the key factors that contribute to WF during the whole life cycle. Moreover, the environmental fate of pollutants among various media (i.e., water, soil, and atmosphere), exposure analyses,

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and toxic effects on humans and the environment are ignored. The International Organization for Standardization (ISO) proposed the life cycle assessment (LCA) method as an international standard for WF analyses to address these problems (ISO 14046, 2014). LCA is a powerful method that quantifies various environmental effects associated with the entire life cycle of certain products, processes, or activities (Finnveden et al., 2009). Currently, most life cycle WF analysis is oriented toward the impact of water scarcity on human health and ecosystem (Hoekstra, 2016; Berger et al., 2014; Boulay et al., 2011). However, the toxicity impact which is related to the changes in water quality also plays indispensable roles in WF evaluation (ISO 14046, 2014; Ridoutt and Pfister, 2013). The limited research based on toxicity impact uses the LCA model (e.g., ReCiPe model) directly (Ridoutt and Pfister, 2013). However, direct application of the LCA model to WF analyses produces inaccurate result (Ma et al., 2018; Bulle et al., 2013). ISO also pointed out that only the influence related to water quality should be considered in WF analysis (ISO 14046, 2014). Meanwhile, few studies have focused on the LCA analysis of crude steel production. These LCA studies are meaningful for other environment issues, such as global warming and fossil depletion. Rossi et al. (2017) focused on the life cycle assessment of steel bridge. Bieda (2012) reported the life cycle inventory of Polish steel production, and Burchart-Korol (2013) and Olmez et al. (2016) conducted LCA analysis of Polish and Turkish ISI, respectively. However, the LCA model applied in previous studies (i.e., IMPACT 2002 + and ReCiPe H) is applicable to Europe; hence conducting LCA analysis of China's ISI via above model may lead to inaccurate results (Li et al., 2016a; Chen et al., 2016). Moreover, the limited LCA research in China mainly concerns air emissions (Li et al., 2016b), and no life cycle water footprint analysis of China's crude steel production has been found.

Thus, the present study aims to (1) perform a life cycle WF assessment based on ISO 14046 standard and a LCA analysis of China's crude steel production, (2) quantify gray and blue WFs of ISI at the national level including direct and indirect processes, (3) analyze the key direct and indirect influencing factors in all crude steel production stages in China, and (4) provide useful suggestions for reducing the environmental burdens of ISI.

2. Methodology

2.1. Scope definition

In this study, the functional unit is a ton of steel billet (i.e., crude steel), which is a quantified reference for inventories and results (ISO 14040, 2006). Fig. 1 illustrates system boundary, and Supplementary Information S1 provides detailed information. Molten iron smelting, steelmaking, on-site reuse of various byproducts (e.g., crude oil and blast furnace gases), and raw material production were all considered in this study. All these processes involve transportation, waste disposal, direct waste emissions, and land occupation. Infrastructure was excluded for its minimal contribution to the overall impact (Frischknecht et al., 2005) and the lack of detailed data. Direct WF represents water consumption in the operation stage and on-site waste (i.e., wastewater, waste gas, and solid) emissions, whereas indirect WF includes the water consumed and the waste discharged in the supply chain (e.g., raw materials production, landfill, and transportation). Additionally, steel billets produced by converter steelmaking technology, which is applied in this study, account for approximately 95% of Chinese crude steel output (CISA, 2016).

2.2. Methodology

Gray (i.e., water resources used to assimilate various pollutants), blue (i.e., consumptive use of surface and groundwater), and green (i.e., rainwater insofar as it does not become run-off) WFs are generally involved in WF analyses (Hoekstra et al., 2011). However, green WF was excluded in this study because rainwater is generally ignored in industrial activities (Berger et al., 2012). The LCA model used in this study (Fig. 2) is a combination of IMPACTWorld + model (Plouffe et al., 2012), IPCC report (IPCC, 2014), and ReCiPe model (Huijbregts et al., 2017). Meanwhile, updated characterization factors of carcinogens, non-carcinogens, and freshwater ecotoxicity was conducted based on USEtox™ model (Huijbregts et al., 2010) by Li et al. (2016a) and Chen et al. (2016). Different from the quantification of updated characterization factors of the aforementioned categories in LCA analysis, the quantification of characterization factors for WF evaluation considered only the intake routes, exposure pathway, and influence related to water quality (Ma et al., 2018). Detailed calculation process is shown in Supplementary Information S2. Moreover, the characterization factor of the remaining midpoints in gray WF evaluation (i.e., aquatic eutrophication) was similar to that of the LCA model in this study. Blue WF evaluation was conducted according to the investigations reported by Berger et al. (2014). Furthermore, the water press index was used to assess the proportion of WF in China's per capita water resources in different regions (NBSC, 2016). The uncertainty analysis results can be obtained from Supplementary Information S3.

2.3. Data sources

The on-site monitoring data on the operation process (e.g., raw material and energy consumption, water use, land occupation, and waste generation) used in this study were collected from an iron and steel plant located in Shandong Province, China. Its annual steel billet (i.e., crude steel) production capacity is approximately 5 million tons. A life cycle inventory was built and is shown in Table 1. Meanwhile, the detailed information of secondary database is shown in Supplementary

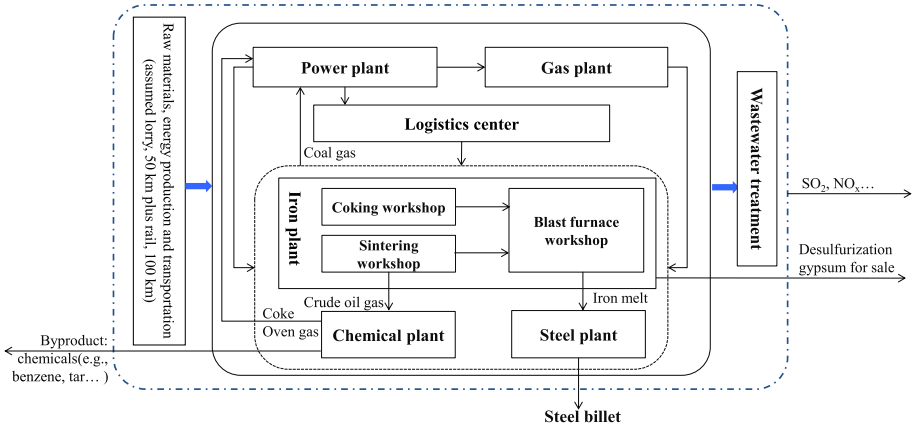


Fig. 1. System boundary.

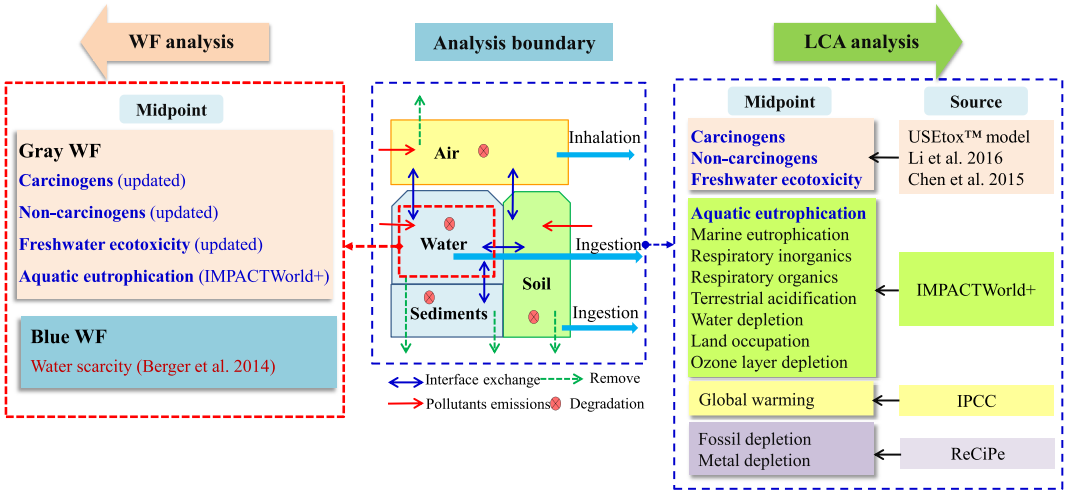


Fig. 2. WF and LCA analysis model.

Table 1
Life cycle inventory. Values are presented per functional unit.

Category	Substance	Amount	Unit
Resource	Mineral	0.02	t
	Dolomite	0.17	t
	Lime	0.27	t
	Refined lime	4.05	kg
	Cobble	4.05	kg
	Ferrosilicon	0.70	kg
	Ferromanganese	2.82	kg
	Silicomanganese	6.90	kg
	Aluminum wire	0.02	kg
	Light burning magnesium ball	4.10	kg
	Electrode	0.23	kg
	Deoxidizer	0.54	kg
	Iron powder	0.36	t
	Rich ore powder	1.32	t
	Coal	0.99	t
	Fresh water	3.58	t
	Electricity	309.89	kWh
	Other ferrous materials	0.36	t
Emissions to air	SO_2	0.93	kg
	PM_{10}	0.15	kg
	NO_x	0.65	kg
	Dioxins	91.32	kg
	Fluoride	3.89	t
Emissions to water	Benzopyrene	3.18	kg
	Sulfide	0.02	g
	Cyanide	0.01	g
	Suspended solids	15.83	g
	Petroleum oil	1.33	g
	Volatile phenol	0.03	g
	NH_3-N	1.12	g
Solid waste	COD	19.10	g
	IHW	0.09	kg
	MSW	0.83	kg

Information S4. In addition, this study aggregated the monitoring data of direct wastewater discharge in 2016 from 31 iron and steel enterprises which are located in the major steel production sites, including Hebei, Jiangsu, Liaoning, Shandong, Shanxi, Hubei, Tianjin, Shanghai, and Gansu provinces (Tables S3–S5). These selected enterprises share approximately 41% of China's total crude steel production, and these enterprises are among the top 100 steel producers in China. Moreover, the monitoring indicators included suspended solid, COD, NH_4-N, phenol, various heavy metals, BOD, and other toxic pollutants.

Table 2
LCA and WF analysis results. Values are presented per functional unit.

Categories	WF		LCA	
	Value	GSD^2	Value	GSD^2
Carcinogens (CTUh)	5.36×10^{-6}	4.36	7.52×10^{-6}	2.49
Non-Carcinogens (CTUh)	7.99×10^{-6}	6.46	2.36×10^{-4}	2.37
Freshwater ecotoxicity (CTUe)	4389.6	1.85	23633.7	2.61
Aquatic eutrophication (kg PO_4^- eq)	9.83×10^{-3}	1.47	9.83×10^{-3}	1.45
Water scarcity (m^3)	7.19	1.11		
Water depletion (m^3)			12.45	1.12
Global warming (kg CO_2 eq)			3776.8	1.13
Fossil depletion (kg oil eq)			657.8	1.25
Metal depletion (kg Fe eq)			10152.2	1.33
Respiratory inorganics (kg $PM_{2.5}$ eq)			2.34	1.15
Respiratory organics (kg NMVOC eq)			2.71	1.24
Terrestrial acidification (kg SO_2 eq)			2.87	1.12
Ozone Layer Depletion (kg CFC-11 eq)			2.99×10^{-6}	1.81
Marine eutrophication (kg N eq)			0.06	1.25
Ionizing radiation (Bq C-14 eq)			179.9	2.65
Land occupation (ha.yr arable)			1.16×10^{-3}	1.45

3. Results and discussion

3.1. WF and LCA analyses results

Table 2 and Fig. 3 present the WF and LCA analyses results. For WF analysis, the potential impact of steel billet production on carcinogens was 5.36×10^{-6} CTUh. The squared geometric standard deviation value (GSD^2) was 4.36. It implies that the results ranged from 1.23×10^{-6} to 2.34×10^{-5} CTUe. Similar analogies for other categories and LCA analysis are presented in Table 2, which also indicates that the LCA scores are significantly higher than the WF scores in aforementioned updated categories (i.e., carcinogens, non-carcinogens, and freshwater ecotoxicity). This finding reveals that the WF will be overestimated by adopting LCA model directly, (Ma et al., 2018; Bulle et al., 2013; Supplementary Information S2). Metal depletion was the key category in LCA analysis primarily because of iron ore consumption, followed by fossil depletion, respiratory inorganics, and global warming (Fig. 3a). The influence of freshwater ecotoxicity and non-carcinogens was relatively small. Meanwhile, the remaining categories played negligible roles. For WF analysis, gray WF (12.04 m^3) provided more obvious contributions than blue WF (7.19 m^3), and indirect

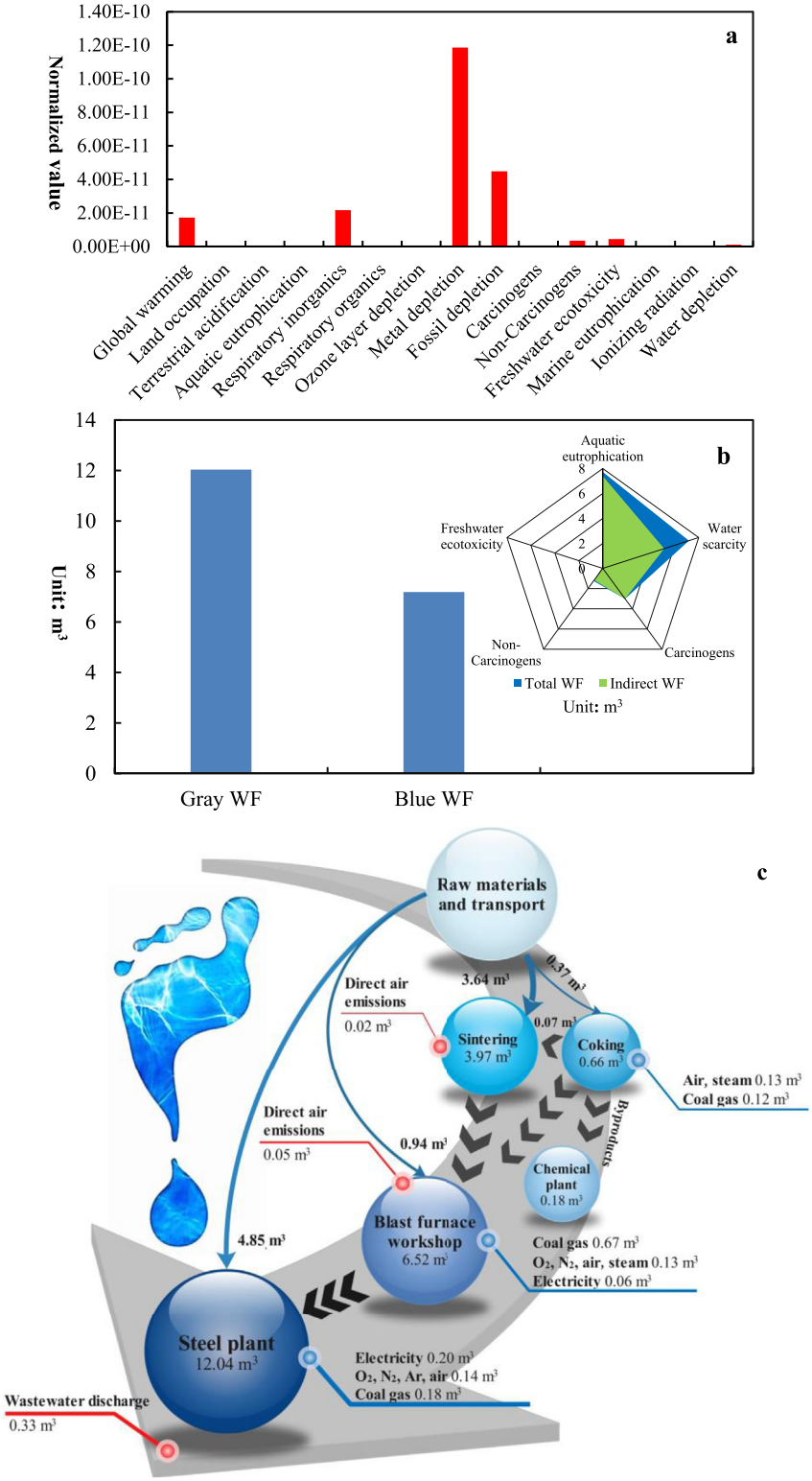


Fig. 3. LCA and WF analysis results: a, normalized midpoints of LCA; b, WF values; c, gray WF of each production unit.

processes played dominant roles in the WF evaluation (Fig. 3b). This result is consistent with previously reported results, including the result indicating that gray WF exceeds blue WF (Gu et al., 2015). Furthermore, the environmental burdens caused by gray WF were mainly derived from aquatic eutrophication, carcinogens, and non-carcinogens. Fig. 3c shows the gray WF of the presented ISI factory. The gray WF of each workshop was mainly from upstream material production, whereas its own energy consumption and wastewater discharge accounted for a small proportion (about 8%). The pollutants emitted into the water through the migration of those emitted to the atmosphere also exerted a slight impact. These results indicate that environmental pollution in the supply chain, rather than direct water consumption, was the main problem in the studied ISI. Additionally, each indicator of the presented production site was lower than the international advanced clean production standard (Table 3), indicating that crude steel production in this study is comparable with the

Table 3
Comparisons with clean production standard of ISI. Values are presented per functional unit.

Consumption/ emission	Freshwater	SO_2	PM_{10}	Iron raw material	Wastewater discharge
Unit	t	kg	kg	t	t
This study	3.58	0.93	0.15	1.07	1.1
International advanced level	4.5	1.0	1.0	1.07	2.0
China's advanced level	8.0	2.0	2.0	1.08	4.0

international advanced level (MEPC, 2006).

3.2. Key factors

The contributions of processes and substances to the main midpoints are shown in Fig. 4. For the common categories (i.e., aquatic eutrophication, carcinogens, and non-carcinogens) in LCA and WF analyses (Fig. 4a), transportation, iron ore mining, coal-based electricity generation, and magnesium oxide production dominated the impact. Ferroalloy production and direct wastewater discharge were responsible for the remaining proportion of aquatic eutrophication. However, compared with the situation in the gray WF evaluation, direct emissions played more effective roles in the LCA analysis. Similar results were obtained for blue WF in which direct freshwater consumption contributed 28.8% to the total value. Meanwhile, the contribution of transportation in gray WF analysis was evident (Fig. 4a). Aside from high environmental burdens, the economic cost of logistics in China's ISI (10.5% of the total cost) is higher than that at the international advanced level which is 6.6% in Japan (CISA, 2016). In this study, transportation was assumed to involve the use of a lorry for solid waste for 50 km distance and a rail for raw materials for 100 km distance. Regardless of the type of transportation (e.g., rail, road, and water), a linear relationship exists between transport distance and its environmental impact (Hong et al., 2015). Therefore, WF can be reduced by selecting a reasonable transportation distance and mode. Among the remaining key categories in the LCA analysis (Fig. 4b), coal mining was the most important factor for fossil depletion, followed by iron ore mining and coal-based electricity generation. The three aforementioned processes contributed 29.0% and 37.8% to global warming and respiratory inorganics, respectively, in which direct emissions and quick lime production played effective roles (i.e., 63.8% for global warming and 39.1% for respiratory inorganics). Transportation and silicomanganese acquisition dominated the influence in the remaining proportion of respiratory inorganics. Additionally, Olmez et al. (2016) and Huang et al. (2010) reported that sintering is the key process of steel production. This result is consistent with the results in Fig. S5.

COD, phosphate, and BOD_5 were responsible for aquatic eutrophication, and they were basically from indirect processes (e.g., iron ore mining, external transport, and ferroalloy production). In addition, direct wastewater emissions contributed 8.5% to the effects of COD. Hexavalent chromium [Cr (VI)] to water generated from indirect emissions of magnesium oxide, iron ore mining, and transportation was the dominant substance for carcinogens in the WF analysis. Meanwhile, mercury (Hg) to air constituted 25.0% of the carcinogens score in LCA analysis. For non-carcinogens in WF analysis, Hg emissions, followed by arsenic (As) to air and titanium (Ti) to soil were the most significant contributing factors. All of these factors were mainly generated through indirect processes, expect for Hg to air, in which 24.8% of its influence originated from direct emissions caused by coal incineration. Moreover, Hg to air dominated the impact in the LCA analysis, given that approximately 94% of the total contribution mainly stemmed from the supply chain. Coal and carbon dioxide played vital roles in fossil depletion and global warming, respectively. Nitrogen oxides, sulfur dioxide, and particulates contributed 51.0%, 33.3%, and 14.8% to respiratory inorganics, respectively.

3.3. Sensitivity analysis

3.3.1. Sensitivity to main contributors

The sensitivity analysis of key factors (Fig. 5) was conducted to analyze the effects of changes in data on the results (ISO 14044, 2006). For gray WF, the efficiency of magnesium oxide production, iron ore mining, and transportation obtained high environmental benefit. The efficiency of ferroalloy, electricity, and direct WF also exhibited a significant influence. Improvements in direct freshwater consumption and iron ore mining played dominant roles in blue WF reduction, followed by transportation, electricity, and ferroalloy. These findings indicate the importance of reducing WF in the supply chain (i.e., raw material production and its transportation). A similar analysis of each key midpoint in WF and LCA analyses can be observed from Fig. 5.

3.3.2. Sensitivity to direct wastewater discharge in gray WF analysis

China's discharge standard for ISI pollutants is mainly concentrated in the atmosphere and involves SO_2, dust, heavy metals, Benz(a)pyrene, and other pollutants (MPI, 2016), especially in blast furnaces (MEPC, 1996a), sintering furnaces (MEPC, 2012a,b), and coke ovens (MEPC, 1996b.) Limited standards for steelmaking wastewater, which focuses on the indicators of COD, pH, suspended solids, NH_4-N, and oils, are applied currently (MEPC, 2012a,b). Accordingly, enterprise self-monitoring indicators of water pollutants show low representativeness. To solve this problem, the aggregated data (Tables S3–S5) was used to assess the gray WF of direct wastewater emissions at national level. The proportion of direct wastewater emissions increased from 1.15% to 5.14%. The lack of enterprise self-monitoring indicators including phosphorus, BOD, Cr (VI), Cu, Zn, Ni, As, and Cd were responsible for this variation. Either the discharge standard of water pollutants or the cleaner production standard ignores the emissions of aforementioned pollutants during crude steel production, except for Zn and As emissions during iron making and sintering, respectively (MEPC, 2012a,b; MEPC, 2006). Thus, abundant controlling indicators of water pollutants should be implemented instead of limited indicators (e.g., COD, pH, and NH_4-N) to avoid the misleading of WF (Maes et al., 2009). These results indicate that the government should systematically and scientifically publish standards and policies based on the actual situation of industries.

3.4. National average WF of ISI

The key factors identified by aforementioned analysis were replaced by national statistics data to assess the national average WF on the basis of Hong et al. (2015) investigation (Supplementary information S6). The national average gray and blue WFs of ISI in China were 1.04×10^{10} m^3 and 5.84×10^9 m^3, respectively. Gray WF contributed more significantly than blue WF did. The key processes of gray WF at national level showed differences when compared with those at the enterprise level (Fig. S7). A significant decrease was observed in iron ore mining (23.4%–14.7%) because of the import (932.51 million ton) and export (60,000 tons) of iron ores (NBSC, 2016). Correspondingly, a significant increasing in direct WF (3.4%–10.9%) and electricity (5.7%–8.3%) and was observed. Missing inventories of wastewater effluents and relatively low electricity consumption and in the reported steel production site were the main reasons. The national direct wastewater discharge of China's ISI was around 717 million tons (CISA, 2016), which accounted for approximately 6% of gray WF caused by national industrial wastewater discharge (NBSC, 2016). However, the gray WF caused by freshwater ecotoxicity was 22.8 times that of the national value. Unreliable national statistics, which are often pressurized by local government (Guan et al., 2012; Liu and Yang, 2009), resulted in the inaccurate evaluation in freshwater ecotoxicity.

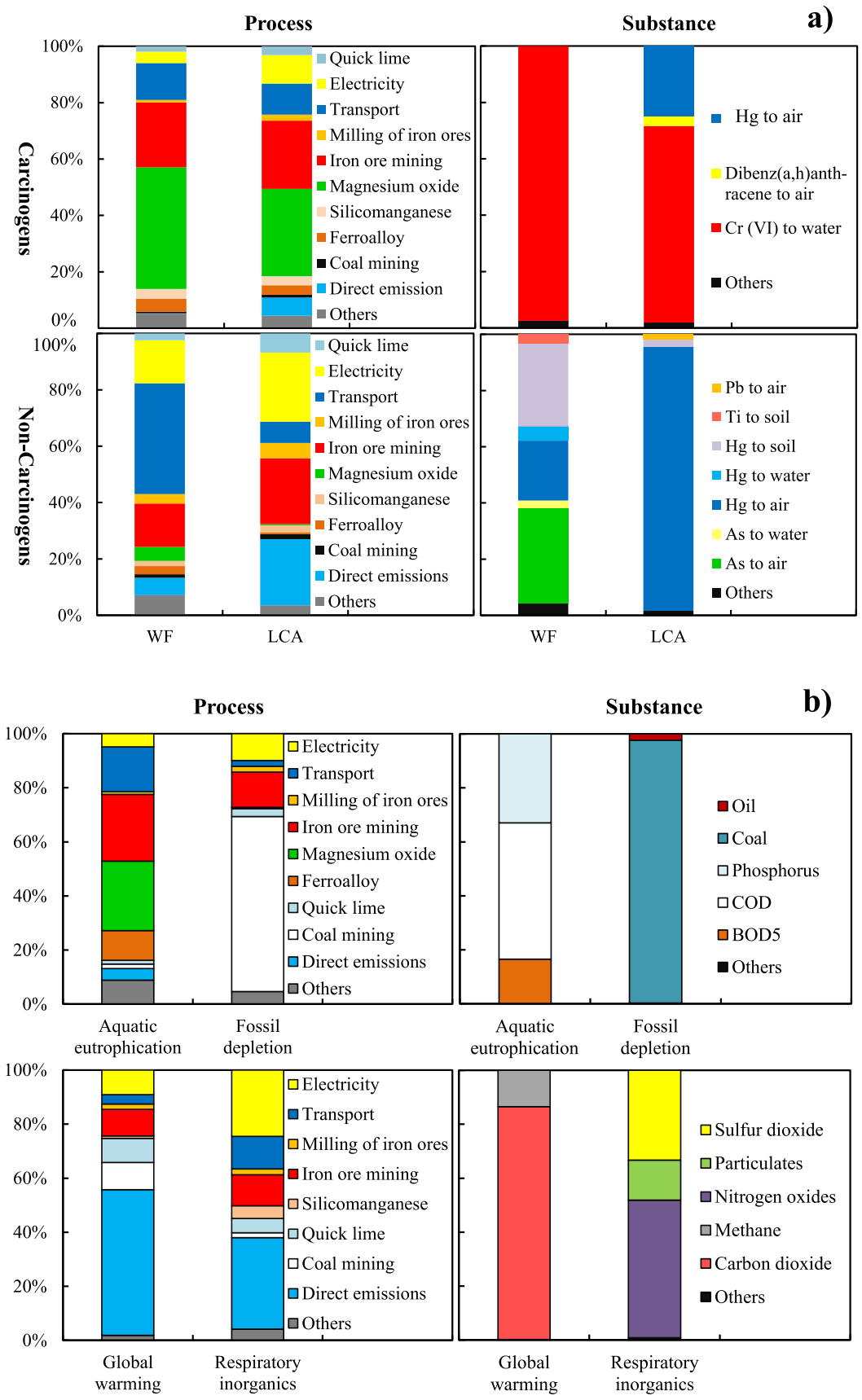


Fig. 4. Contribution of processes and substances to main categories: a) contribution to carcinogens and non-carcinogens via WF and LCA model; b) contribution to main categories in LCA analysis.

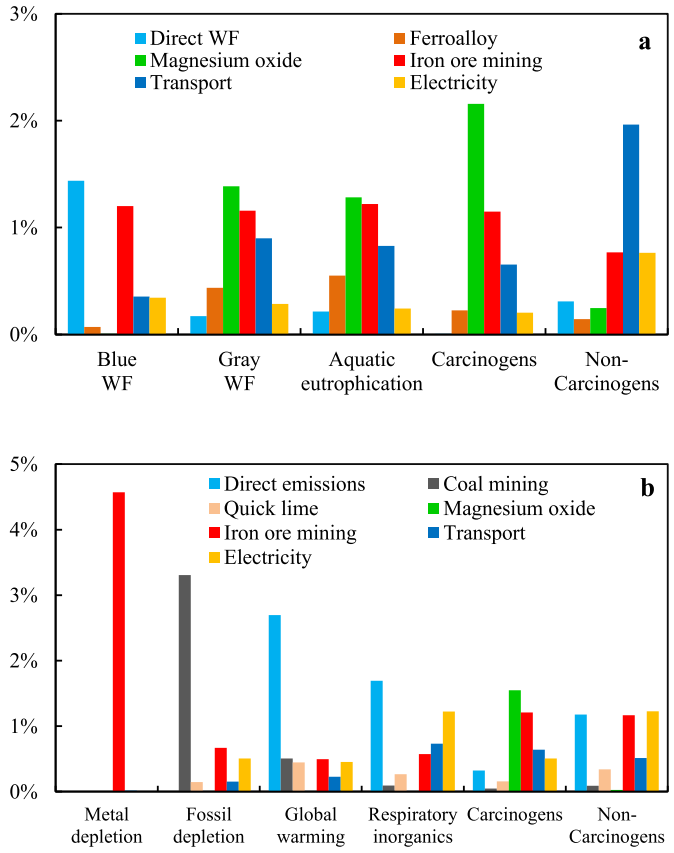


Fig. 5. Sensitivity to main contributors: a, WF analysis; b, LCA analysis.

Meanwhile, Wu (2007) and Guan et al. (2012) pointed out that statistical data are collected via a top-down statistical system in China and generally shows a feature of an opaque collection approach. Table S7 compares the effects of freshwater ecotoxicity using enterprise aggregated data and national statistical data. Numerous indicators were found to be missing in national statistical yearbook (NBSC, 2016). Accordingly, constructing database through a bottom-up statistical approach and further enriching monitoring indicators based on specific characteristics of different industries are highly necessary for water pollution management.

Fig. 6 illustrates the direct water pressure caused by 1 ton of aggregated ISI wastewater emissions, while water resource quality and quantity status in China is shown in Fig. S8 (NBSC, 2016; MEPC, 2016). The water pressure of Ningxia was the most severe, followed by Tianjin, Beijing, Hebei, and Shandong. Notably, the water resources per capita of these provinces were all lower than 500 m^3/capita/year which represented absolute water scarcity (UNESCO, 2015), especially in Beijing and Tianjin where water pollution is prevalent. By contrast, the pressure in South China, where the water resource is relatively abundant and slightly polluted, was low. Thus, the quality and quantity of regional water resources should be considered in the assessment model. Furthermore, other key factors (Fig. 4), such as transportation, iron ores production, and electricity, revealed regional differences (CISA, 2016; NBSC, 2016). Thus, developing ISI in Northwest China (e.g., Ningxia Province), where the transportation burden is high, iron ores and water resources are short, and energy structure is dominated by coal power, is thoughtless (CISA, 2016; NBSC, 2016). Reasonable layout of ISI is a key factor in reducing WF, and ISI can be located in areas with low water pressure (e.g., Anhui Province).

The LCA results of crude steel production in Europe (ID number: EI3ADUNI000061667502562; Ecoinvent Centre, 2015), Poland (Burchart-Korol, 2013), and the United States (ID number: USLCI-XXX000006578700334; NREL U.S, 2014) obtained via the ReCiPe model were compared with the results of China's crude steel production (Table 4). Data on Europe, Poland, and the US represent European mixed technology in 2001, Polish steel industry in 2011, and US and Canadian average technologies based on input-output table in 2013, respectively. The environmental burden caused by water depletion in China was lower than that in the three aforementioned areas due chiefly to less direct freshwater consumption. The environmental burden derived from climate change, fossil depletion, and human toxicity in China was higher than that in the US and lower than that in Europe and Poland. Metal depletion in China was significantly lower than that in Europe, and Poland's value is 12.2% of that for China. Overall, the technical level of the presented steel production site was higher than that of Europe and slightly lower than that of Poland and the US. These results indicate that steelmaking technology has improved significantly over time. The inventory database should be updated constantly to represent the current technological level effectively. Notably, the power involved in this study was coal-based electricity, which accounts for approximately 76% of China's power grid (NBSC, 2016). Other types of electricity also play important roles in China, especially hydropower which accounts for about 20% of China's electricity composition (NBSC, 2016). If hydropower whose environmental burden is lighter than coal power (Ma et al., 2017) is considered in this study and national electricity consumption (467.9 KWh/t-crude steel) is further reduced to European levels of 132.5 KWh/t-crude steel

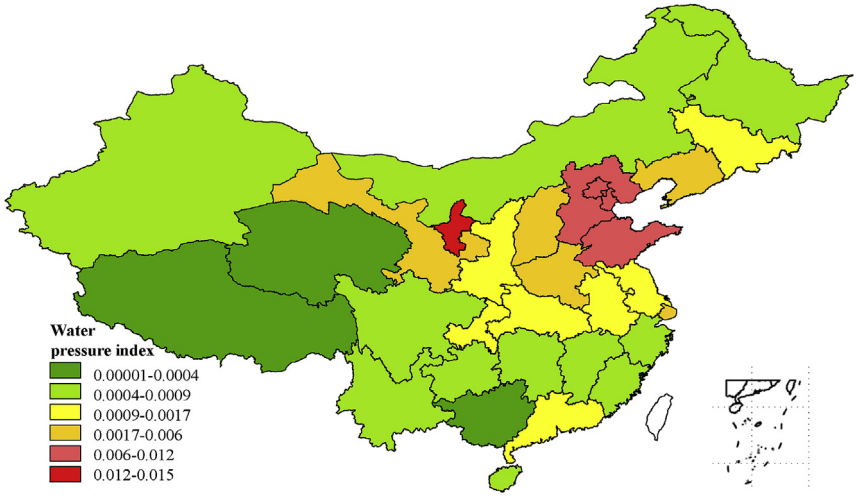


Fig. 6. Provincial water pressure status of 1 ton aggregated ISI direct wastewater discharge in China.

Table 4
Comparison of steel production. Values are presented per functional unit.

Categories	Unit	China	Europe	Poland	US
Climate change	kg CO_2 eq	2175.6	4866	1703	2044
Water depletion	m^3	12.3	47.6	87.4	–
Metal depletion	kg Fe eq	6965.6	13846.3	850	–
Fossil depletion	kg oil eq	646.7	1155.9	529	516.6
Human toxicity	kg 1,4-DB eq	94.7	2326.6	643	39.5

(Burchart-Korol, 2013), then the gray WF in this study will decrease by 6.6%. The analysis indicates that WF can be reduced by optimizing energy consumption efficiency especially in sintering (Fig. S5) and adopting clean energy.

4. Conclusions

This study performed LCA and WF analyses of ISI in China and explored the key factors that contribute to environmental burdens. The LCA analysis showed that the environmental degeneration caused by crude steel production was mainly derived from metal depletion, fossil depletion, respiratory inorganics, global warming, freshwater ecotoxicity, and non-carcinogens. In this study, 1 ton of crude steel production obtained 12.60 m^3 of gray WF at the national level in China, and the score was 1.8 times that of blue WF (7.09 m^3). Compared with the results of the LCA analysis, the WF midpoints had lower scores in carcinogens, non-carcinogens, and freshwater ecotoxicity categories. This result indicates that the life cycle impact assessment model cannot be directly applied in WF analysis because LCA involves intake routes and exposure pathways unrelated to water quality (Fig. S3). The most important contributor to gray WF was aquatic eutrophication, whose effects were dominated by indirect processes (95.7%), such as iron ore mining, raw material transportation, and electricity generation, which were also the key factors in LCA analysis. Direct emissions also played important roles in global warming and respiratory inorganics. Improvement of these processes would reduce the environmental burden. COD, Cr (VI) to water, phosphate to water, BOD_5, Hg to air and soil, and As to air were the key substances for gray WF. Meanwhile, Hg emissions provided a large contribution in LCA analysis (e.g., 25.0% for carcinogens and 96.5% for non-carcinogens). Moreover, nitrogen oxides, particulates, and sulfur dioxide were the main substances for respiratory inorganics, whereas carbon dioxide dominantly influenced global warming. Additionally, water stress caused by direct wastewater discharge varied at the provincial level, indicating that the WF analysis model should consider regional variations in the quantity and quality of water resources. Furthermore, numerous indicators, including COD, SS, NH_4-N, oils, phosphorus, BOD, Cr (VI), Cu, Zn, Ni, As, and Cd, should be implemented for controlling pollutant emissions.

This study provides valuable information for water resource management, enterprise layout, and scientific pollution control of ISI. Moreover, this study is meaningful for understanding the differences between WF and LCA analyses, and it can provide some useful ideas about how the WF is appropriately conducted via LCA model. However, some limitations still existed in this study. Although converter technology, which is widely used in China, was analyzed in this study, other technologies should also be investigated further. Moreover, studies on dynamic database and spatial disparity in national WF evaluation and sustainable WF assessments, including economic and social, should be involved. Our research group will address these limitations in our future work.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jenvman.2018.07.027.

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